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CSC 125 — Past Questions & Materials
by Renance

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Paper 1 of 1 — CSC 125

Level 100 • PostUTME • Handout

What are the requirements and specifications of embedded system?

Embedded systems must be designed to fit on a single chip, and perform fast enough to process that data in real time, furthermore, they must also be designed in a way to consume the least power possible to be able to extend battery life.

2 Embedded System Design

Process

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In this lecture various steps involved in Embedded system design process will be discussed.

1. Design process steps:

There are different steps involved in Embedded system design process. These steps depend on the design methodology. Design methodology is important for optimizing performance, and developing computer aided design tools. It also makes communication between team members easier. Figure1 shows the steps in the design process. They are requirements gathering, specification formulation, architecture design, building of components, and system integration.

The steps in the design process can be viewed as top down view and bottom up view. Top-down view begins with the most abstract description of the system and concludes with concrete details. Bottom-up view starts with components to build a system. Bottom-up design steps are shown in the figure as dashed-line arrows. We need bottom-up design because we do not have perfect insight into how later stages of the design process will turn out.

The major goals of the design to be considered are :

- Manufacturing cost;
- Performance (both overall speed and deadlines) and
- Power consumption.

The tasks which need to be performed at each step are the following.

- We must analyze the design at each step to determine how we can meet the specifications.
- We must then refine the design to add details.
- We must verify the design to ensure that it still meets all system goals, such as cost, speed, and so on.

We will discuss each step of the design process in detail.

1.1 Requirements

Informal descriptions gathered from the customer are known as requirements.

The requirements are refined into a specification to begin the designing of the system architecture. Requirements can be functional or non-functional requirements. Functional requirements need output as a function of input. Non-functional requirements includes performance, cost, physical size, weight, and power consumption. Performance may be a combination of soft performance metrics such as approximate time to perform a user-level function and hard deadlines by which a particular operation must be completed. Cost includes the manufacturing, nonrecurring engineering(NRE) and other costs of designing the system. Physical size and weight are the physical aspects of the final system. These can vary greatly depending upon the application. Power consumption can be specified in the requirements stage in terms of battery life.

Example:

Let us consider an example of a GPS system. The sample Requirement Form is shown in Figure 2.

1.2 Specification

Requirements gathered is refined into a specification. Specification serves as the contract between the customers and the architects. Specification is essential to create working systems with a minimum of designer effort. It must be specific, understandable and accurately reflect the customer's requirements.

Example:

Considering the example of the GPS system, the specification would include details for several components:

- Data received from the GPS satellite constellation
- Map data
- User interface
- Operations that must be performed to satisfy customer requests
- Background actions

1.3 Architecture Design

The specification describes only the functions of the system. Implementation of the system is described by the Architecture. The architecture is a plan for the overall structure of the system. It will be used later to design the components. The architecture will be illustrated using block diagrams as shown below.

Example:

A basic block diagram of the GPS system shows the major operations and the data flow among the blocks.

This block diagram(figure 3) is an initial architecture that is not based either on hardware or on software but combination of both. This block diagram explains about GPS navigating system where GPS receiver gets current position and the

destination is taken from user, digital map for source to destination is found from database and displayed by the renderer. The system block diagram may be refined into two block diagrams – hardware and software.

1.3.1 Hardware block diagram :

Hardware consists of one central CPU surrounded by memory and I/O devices. We have chosen to use two memories that is frame buffer for the pixels to be displayed and separate program/data memory for general use by the CPU. The GPS receiver is used to get the GPS coordinates, and the panel I/O is used to get the destination from the user.

1.3.2 Software block diagram :

Figure 5. GPS system software

The software block diagram closely follows the system block diagram. We have added a timer to control when we read the buttons on the user interface and render data onto the screen.

To have a truly complete architectural description, we require more details, such as where units in the software block diagram will be executed in the hardware block diagram and when the operations will be performed in time.

Architectural descriptions must be designed to satisfy the functional and non-functional requirements. Not only must all the required functions be present, but we must meet cost, speed, power and other non-functional constraints. Starting out with a system architecture and refining that to hardware and software architectures is one good way to ensure that we meet all specifications. We can concentrate on the functional elements in the system block diagram, and then consider the non-functional constraints when creating the hardware and software architectures.

How do we know that our hardware and software architectures in fact meet constraints on speed, cost, and so on?

- Estimate the properties of the components in the block diagrams (Example: search and rendering functions in the moving map system)
- Accurate estimation derives in part from experience, both general design and particular experience.
- All the non-functional constraints are estimated. If the decisions are based on bad data, those results will show up only during the final phases of design.

1.4 Hardware and Software components

The architectural description tells us what components we need. The component design effort builds those components in conformance to the architecture and specification. The components in general includes both hardware and software modules. Some of the components will be ready-made (example :CPU, memory chips).

Example:

In the moving map, GPS receiver is a predesigned standard hardware component. Topographic software is a standard software module which uses standard routines to access the database. Printed circuit board are the components which needs to be designed. Lots of custom programming is required.

When creating these embedded software modules, ensure the system runs properly in real time and that it does not take up more memory space than allowed. The power consumption of the moving map software example is particularly important. You may need to be very careful about how you read and write memory to minimize power. For example, memory transactions must be carefully planned to avoid reading the same data several times, since memory accesses are a major source of power consumption.

1.5 System integration

After the components are built, they are integrated. Bugs are typically found during the system integration. Good planning can help us to find the bugs quickly. By debugging a few modules at a time, simple bugs can be uncovered. By fixing the simple bugs early, more complex or obscure bugs can be uncovered. System integration is difficult because it usually uncovers problems. The debugging facilities for embedded systems are usually much more limited than the desktop systems. Careful attention is needed to insert appropriate debugging facilities during design which can help to ease system integration problems.

2. Summary

In the Embedded system design process, information is first collected and refined in the Requirement step. Specification uses the refined information to describe the functions of the system which accurately reflects the customer's requirements and also serves as the contract between the customer and the designer. The functions described by the specification are implemented by the Architecture design. The architectural design describes the components we need which will include both the hardware and software components. After the components are built, they are integrated to get the final system.

What is a solderless circuit board?

The solderless breadboard (generally a large, white, plastic component with rows and columns of holes) provides a working space where temporary circuits can easily be built¹. Leads of electrical components (e.g. resistors) can easily be pushed into the breadboard holes.

What is prototyping in breadboard?

What does it mean? It means that, when you want something quickly made (or you are just testing/learning and making a prototype), there is no need to use a soldering iron, you just plug each component in the right place (if you are following a tutorial of course) and it works!

What is the introduction of breadboard?

A breadboard, solderless breadboard, or protoboard is a construction base used to build semi-permanent prototypes of electronic circuits. Unlike a perfboard or stripboard, breadboards do not require soldering or destruction of tracks and are hence reusable.

What is circuit prototyping?

In electronics, prototyping means building an actual circuit to a theoretical design to verify that it works, and to provide a physical platform for debugging it if it does not.

What are the different types of solderless connectors?

Solderless Terminals Terminals and Connectors

Some of the offerings are; Ring terminal, Spade terminal, Flanged Spade Terminal, Hi-Temp Terminal, Spring Spade Terminal, Butt Connectors, Female Disconnects, Male Disconnects, Flag Disconnects, Wire Nuts and Closed End Connectors.

What are the two types of circuit boards?

What are the Different Types of PCBs?

□ Single Sided PCBs. Single sided PCBs are the basic type of circuit boards, which contain only one layer of substrate or base material. ...

□ Double Sided PCBs. ...

□ Multi-layer PCBs. ...

□ Rigid PCBs. ...

□ Flexible PCBs. ...

□ Rigid-Flex-PCBs. ...

□ High-Frequency PCBs. ...

□ Aluminum backed PCBs.

What is prototyping with example?

Prototyping is a creative and brainstorming process that allows for coming up with effective suggestions and validating your hypothesis with the help of future users.

The only way to develop a solution the users will enjoy is to fit their expectations and match their preferences.

What is called prototyping?

The prototyping model is a systems development method in which a prototype is built, tested and then reworked as necessary until an acceptable outcome is achieved from which the complete system or product can be developed.

What are the two main types of prototyping?

What are the different types of prototyping? The 4 types of project prototyping are feasibility prototypes, low-fidelity user prototypes, high-fidelity user prototypes and live-data prototypes.15 Jul 2024

What are the two types of breadboard?

There are two major types of breadboards; these are solder and solderless boards. Solder boards are boards you have to solder components onto (per the name).

These are most of your standard circuit boards, and if you flip one over you'll notice that all of the connections are soldered to the board itself.

What is breadboard circuit diagram?

Unlike a circuit diagram or a schematic (which use symbols to represent electronic components; see the Advanced section to learn more), breadboard diagrams make it easy for beginners to follow instructions to build a circuit because they are designed to look like the "real thing."

What is the basic use of breadboard?

The purpose of the breadboard is to make quick electrical connections between components- like resistors, LEDs, capacitors, etc- so that you can test your circuit before permanently soldering it together. Breadboards have many small sockets on them, and some groups of sockets are electrically connected to each other.

What is a prototype board?

Perf boards (or breadboards) are a convenient platform when prototyping electronic circuits. They're a quick, easy way to mount and connect leaded electronic components together. They're also easily interchangeable with testing all new configurations.16 Mar 2020

What is the difference between a breadboard and a prototype board?

The major differences are: Soldered protoboards use solder, while the solderless doesn't. In terms of performance, the soldered breadboards are high-performing, due to the permanent capture of the leads. In terms of versatility, you will find the solderless protoboards a better option.

How do you make a prototype circuit?

How To Prototype Circuit Boards

1. Step 1: Designing the Schematic. ...
2. Step 2: Creating the PCB Layout. ...
3. Step 3: Generating the Files. ...
4. Step 4: Choosing the Prototyping Method. ...
5. Step 5: Manufacturing the PCB. ...
6. Step 6: Drilling the Holes. ...
7. Step 7: Soldering the Components. ...
8. Step 8: Testing the Prototype Circuit Board.

What are the 5 main types of circuits?

There are five types of electric circuit:

- ☐ Open Circuit.
- ☐ Closed Circuit.
- ☐ Short Circuit.
- ☐ Series Circuit and.
- ☐ Parallel Circuit.

How many circuit boards are there?

There are 6 main types of printed circuit boards, named after their number of layers and flexibility: Layered: Single-sided PCBs – Basic circuit board with all components on one side comprising of a single conductive copper layer on the substrate and then the solder mask.20 Jul 2022

What are circuit boards called?

A printed circuit board, or PC board, or PCB, is a non-conductive material with conductive lines printed or etched. Electronic components are mounted on the board and the traces connect the components together to form a working circuit or assembly.

What is the most common solderless connector?

Looped-end connectors

Looped-end connectors

The most common solderless connection has the looped end of a wire (Figure 9.3. 1) held in place by a set-screw at an electrical terminal (Figure 9). Note that the direction of the loop is the same as the direction the screw is turned when it is tightened (clockwise).11 Sept 2021

Which device is used for solderless connection?

Solderless breadboard

It's the most used device when creating temporary circuits. It is called solderless because no soldering is required, you can just plug in the components. A component can easily be removed from a breadboard if you make a mistake, or when starting a new project.12 May 2023

How to use solderless connectors?

What are two types of prototyping?

What are the Types of Prototypes in Design Thinking?

□ Low-Fidelity Prototypes: These are basic and often quick to produce. ...

□ High-Fidelity Prototypes: More sophisticated and detailed, this is one of the types of prototypes that closely resembles the final product in aesthetics, functionality, and interaction.

What are 3 prototypes?

Prototypes refer to the simulations/samples/proofs of concept with varied fidelity levels - low, medium and high depending on how interactive it is.24 Jan 2022

What is another word for prototype?

prototype (noun as in original, example) Strongest matches. mock-up model precursor. Strong matches. ancestor antecedent archetype criterion first forerunner ideal norm paradigm pattern precedent predecessor standard type.

What is the purpose of prototyping?

Prototyping is an experimental process where design teams implement ideas into tangible forms from paper to digital. Teams build prototypes of varying degrees of

fidelity to capture design concepts and test on users. With prototypes, you can refine and validate your designs so your brand can release the right products.

What is the difference between breadboard and solderless breadboard?

Is one better than the other, or is it just personal preference? The main difference between solderless and soldered breadboards is that solderless breadboards use spring-loaded clips to connect components, while soldered breadboards use solder to connect components.26 Jun 2023

Four types of Prototypes: Role, Look and feel, Implementation, and Integration.

Prototyping is an activity core to designing and engineering, though an activity that has traditionally been under examined.

What are 2 advantages of prototyping?

Benefits of Prototyping

- Identify design flaws early on in the development process.
- Reduce the costs of product development.
- Test functionality and performance of a product.
- Receive feedback from stakeholders and end-users.
- Improve communication between stakeholders and design teams.

What are the 3 forms of prototypes?

Prototypes come in many different shapes and sizes. You can elaborate your prototype on or with paper, elaborate it digitally or, for example, have it printed with a 3D printer. The different types of prototypes are classified into three different categories: low fidelity, medium fidelity and high fidelity.14 Mar 2023

Why use breadboard for prototyping?

Breadboards are used most commonly in prototyping applications. The fact that solderless breadboards don't require circuitry components to be affixed semi-permanently to the surface of a PCB makes it much easier and quicker to manoeuvre and swap them until you achieve the desired effect.

Installation instructions

1. Download the latest release (The download will start after you click this link. ...
2. Double-click the executable (.exe) file.
3. Follow the instructions in the installation guide.
4. When completing the setup, leave Run Arduino IDE ticked to launch the application, or launch it later from the Start Menu.

How do I install Arduino IDE?

How to Install Arduino IDE on Windows 10

1. Step 1: Go to the Arduino.cc Website. ...
2. Step 2: Click on the Download Link. ...
3. Step 3: Begin the Download. ...
4. Step 4: Begin the Installation Process. ...
5. Step 5: Select What to Install. ...

6. Step 6: Choose the Installation Path. ...

7. Step 7: Finish the Installation. ...

8. Step 8: Launch the Arduino IDE.

What is the Arduino IDE?

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them.17 Jan 2024

What is setup in Arduino IDE?

Description. The setup() function is called when a sketch starts. Use it to initialize variables, pin modes, start using libraries, etc. The setup() function will only run once, after each powerup or reset of the Arduino board.

What is the system requirement for Arduino IDE?

For Windows 10, Java version 8u51 is required to run. For Macs, it requires OS X version 10.8. 3 or above. The Arduino website states that any Linux flavor should work and that only the architecture (Intel 32 bit, Intel 64 bit, ARM) matters [17].

What is Arduino used for?

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online.25 Jan 2022

How to install Arduino IDE theme?

zip packaged theme for Arduino IDE 1.

1. Open the Sketchbook folder. ...
2. If open, close Arduino IDE.
3. In the sketchbook folder, create a subfolder named theme .
4. Drag and drop the custom theme
5. Open the Arduino IDE, select the theme from the Arduino IDE at File > Preferences > Theme menu.
6. Restart the Arduino IDE again.

3 Jun 2024

Where is Arduino IDE used?

The Arduino IDE (Integrated Development Environment) is used to write the computer code and upload this code to the physical board. The Arduino IDE is very simple and this simplicity is probably one of the main reason Arduino became so popular.

What is Arduino IDE function?

The Arduino IDE is an open-source software, which is used to write and upload code to the Arduino boards. The IDE application is suitable for different operating systems such as Windows, Mac OS X, and Linux. It supports the programming languages C

and C++. Here, IDE stands for Integrated Development Environment.

Where are Arduino IDE examples?

Built-in Examples are sketches included in the Arduino Software (IDE), to open them click on the toolbar menu: File > Examples.

How to program Arduino with IDE?

Explore a step-by-step guide to setup the environment of Arduino programming

1. Download & install the Arduino environment (IDE) ...
2. Launch the Arduino IDE. ...
3. If needed, install the drivers. ...
4. Connect the board to your computer via the USB cable. ...
5. Select your board. ...
6. Select your serial port. ...
7. Open the blink example. ...
8. Upload the program.

What language is used on Arduino IDE?

C++ programming language

Arduino uses a variant of the C++ programming language. The code is written in C++ with an addition of special methods and functions. Moreover, when you create a 'sketch' (the name given to code files in this language), it is processed and compiled to machine language.25 Jan 2023

What is the full meaning of Arduino IDE?

Arduino Integrated Development Environment

Description: Arduino Integrated Development Environment (IDE) is an open source IDE that allows users to write code and upload it to any Arduino board. Arduino IDE is written in Java and is compatible with Windows, macOS and Linux operating systems.

What are the steps to install Arduino IDE?

Installation instructions

1. Download the latest release (The download will start after you click this link. ...
2. Double-click the executable (.exe) file.
3. Follow the instructions in the installation guide.
4. When completing the setup, leave Run Arduino IDE ticked to launch the application, or launch it later from the Start Menu.

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What is the main function in Arduino?

When you run a "standard" C/C++ program, you have to write a "main" function. This main function will be called first, and from there, you will call other functions and execute the functionalities of your program. In Arduino, there is no main function. This is replaced by setup and loop.

What are the advantages of Arduino?

The advantages of using Arduino include low cost, portability, light weight, low power consumption, ease of use, and the ability to combine with various electronic products. Arduino can be a useful tool in various industries such as electronics, libraries, and mining.

What is the application of Arduino IDE?

The Arduino Software (IDE) makes it easy to write code and upload it to the board offline. We recommend it for users with poor or no internet connection. This software can be used with any Arduino board. There are currently two versions of the Arduino IDE, one is the IDE 1.16 Jan 2024

SERIAL COMMUNICATION.

What is a serial communication?

What is serial communication? Serial communication is a communication method that uses one or two transmission lines to send and receive data, and that data is continuously sent and received one bit at a time.

What is serial communication in Arduino?

Introduction. Serial communications provide an easy and flexible way for your Arduino board to interact with your computer and other devices. This chapter explains how to send and receive information using this capability. Chapter 1 described how to connect the Arduino serial port to your computer to upload sketches.

What is the theory of serial communication?

Serial transmission involves sending one bit at a time, such that the data is spread out over time. The total number of bits transmitted per second is called the baud rate. The reciprocal of the baud rate is the bit time, which is the time to send one bit. Most microcontrollers have at least one UART.

What are the two types of serial communication?

There are two broad types of serial communication: synchronous and asynchronous. There are a very large number of different standards and protocols for serial communication, ranging from the very simple to the seriously complicated. It is important to match the right protocol with the right application.

What are the examples of serial communication?

To reduce the number of pins in a package, many ICs use a serial bus to transfer data when speed is not important. Some examples of such low-cost lower-speed serial buses include RS-232, DALI, SPI, CAN bus, I²C, UNI/O, and 1-Wire. Higher-speed serial buses include USB, SATA and PCI Express.

What is serial communication and parallel communication with example?

One such example is PCI Express. Unlike parallel communication which sends multiple bits at the same time, serial communication sends them one by one. It can use as little as one wire, but two or more are most common. It usually never uses more than four.29 Mar 2023

What are the serial communication protocol devices?

Serial communication protocols define how data is transmitted sequentially between devices over a communication link based on standardized rules and conventions. Common serial communication protocols include RS-232, SPI, USB, CAN, I2C, and Modbus.11 Mar 2024

Is USB serial communication?

USB has become the preferred serial communication protocol between digital products like computers and peripheral devices as it can transmit data across longer cables and offers higher transmission rates.26 Jan 2022

Is serial communication analog or digital?

Essentially, serial communication uses digital inputs and outputs to send strings of ones and zeros that represent analog and digital values. They are often used in favor of individual analog and/or digital signals that would require many individual wires as well as many input and output points.10 Jul 2019

What are the advantages of serial computing?

Serial mode offers the advantage of fewer traces on the pc board, and fewer pins on the devices. Parallel offers the advantage of transferring 8 data bits per I/O clock cycle, but at the disadvantage of many more pins required on the devices.

What is meant by serial communication?

Serial communication is a simple means of sending data to long distances quickly and reliably. The most commonly used serial communication method is based on the RS232 standard.

Is USB serial or parallel?

What is serial communication basic knowledge?

What are the advantages of serial communication?

Serial communication, characterized by its sequential transmission of bits, offers benefits in terms of simplicity, reliability, and cost-effectiveness, especially over longer distances. The slower speeds, however, may be a drawback in applications demanding rapid data transfer.14 Aug 2023

Is Bluetooth a serial communication?

Serial Communication -- Bluetooth is like a RF version of serial communication.

What is Pandora's Box project?

- A photographic portrait series documenting and celebrating the changing face and profound cultural influence of womxn artists and art practitioners (inclusive of trans; non-binary; gender-queer, and female individuals) over the past six decades.
- A quietly radical cultural intervention.

SENSORS AND ACTUATORS

What is a sensor and actuator?

A sensor monitors conditions and signals when changes occur. An actuator receives a signal and performs an action, often in the form of movement in a mechanical

machine.

What are smart sensors and actuators?

A smart sensor is a device that takes input from the physical environment, like a flow or pressure, and uses its built-in computing resources to perform predefined functions upon detection of specific input and then process data before passing it on.

Smart actuators are similar in their capabilities to smart sensors.2 Jul 2024

Smart Sensors & Actuators: Basics and Benefits

What is the purpose of sensors actuators and smart objects?

Sensors are used to sense the environment and actuators are used to affect or change the environment. The sensors and actuators attached to a smart object range from very simple to very complex. A smart object that measures the temperature needs only a simple temperature sensor.

What is the conclusion of sensor and actuator?

What actuator means?

Defined simply, an actuator is a device that converts energy, which may be electric, hydraulic, pneumatic, etc., to mechanical in such a way that it can be controlled. The quantity and the nature of input depend on the kind of energy to be converted and the function of the actuator.

What are 4 actuators examples?

Common examples of actuators include electric motors, stepper motors, jackscrews, electric muscular stimulators in robots, etc.

What is meant by sensor?

A sensor is a device that detects and responds to some type of input from the physical environment. The input can be light, heat, motion, moisture, pressure or any number of other environmental phenomena.

What are the different types of sensors?

Types of Sensors

- Position Sensors.
- Pressure Sensors.
- Temperature Sensors.
- Force Sensors.
- Vibration Sensors.
- Piezo Sensors.
- Fluid Property Sensors.
- Humidity Sensors.

How does a sensor work?

A sensor converts the physical action to be measured into an electrical equivalent and processes it so that the electrical signals can be easily sent and further processed. The sensor can output whether an object is present or not present

(binary) or what measurement value has been reached (analog or digital).

What are the disadvantages of sensors?

Disadvantages of Sensors:

- Sensors may be expensive, in particular, if they're excessive-precision or specialized.
- Sensors may be fragile and can want to be covered from bodily harm or severe temperatures.
- Sensors may be tormented by interference from different sources, electromagnetic fields, or sensors.

More items...

What are smart sensors used for?

Smart sensors are used in many sectors to meet challenges with innovative solutions. In industry, they are used for equipment monitoring, quality control, and process production optimization. Because of their ability to detect anomalies and prevent breakdowns, they help to reduce downtime and improve productivity. 17 Jun 2024

Key Differences between Sensors and Actuators

A sensor is a device that detects changes or events in the environment and transmits that data to other electronic devices. In contrast, an actuator is a machine component that moves and controls mechanisms. Electrical signals are generated via sensors.

What is the main function of an actuator?

An actuator is a component of a machine that produces force, torque, or displacement, usually in a controlled way, when an electrical, pneumatic or hydraulic input is supplied to it in a system (called an actuating system). An actuator converts such an input signal into the required form of mechanical energy.

What device is an actuator?

An actuator is a device that converts energy into some kind of "operation", such as linear movement, rotation, or bending. In addition to electricity, the input energy comes in a variety of forms, such as air (pneumatic) or oil (hydraulic) pressure, magnetic force from electromagnets, steam, and heat.

Is a LED an actuator?

However simple these leds may look, these are actually binary actuators. Note that you may dim a led by using Pulse Width Modulation (PWM). In that case it becomes an analog actuator.

Is a LED an actuator?

Are actuators AC or DC?

Electric linear actuators are driven by either an AC or DC motor. This article will highlight the main differences between these two motor types in order to assist you

in the selection of the correct linear actuator motor. Want to convert between AC and DC? Our PA-20 has an AC inverter! 1 Mar 2018

What is an actuator in ICT?

An actuator is the mechanism by which a control system acts upon an environment. The control system can be simple (a fixed mechanical or electronic system), software-based (e.g., a printer driver, robot control system), or a human or other agent.

What are light sensors used for?

Definition. Light sensors are a type of photodetector (also called photosensors) that detect light. Different types of light sensors can be used to measure illuminance, respond to changes in the amount of light received, or convert light to electricity.

What is sensor types?

There are two types of sensors: Direct Sensor: A sensor that can convert a non-electrical stimulus into an electrical signal with intermediate stages. Eg: thermocouple (temperature to voltage). Indirect Sensor: A sensor that multiple conversion steps to transform the measured signal into an electrical signal.

PANDORA'S BOXX PROJECT

What is Pandora's box trying to explain?

Pandora's box is a metaphor for something that brings about great troubles or misfortune, but also holds hope. In Greek mythology, Pandora's box was a gift from the gods to Pandora, the first woman on Earth. It contained all the evils of the world, which were released when Pandora opened the box.

How to make a Pandora's box?

Cover your workspace with newspaper and paint the outside of a shoebox a hopeful color like yellow, gold, or pink. Paint the inside of the box a dark red or black to represent the evil inside. Then, make the outside enticing by gluing on jewels and glitter.

Who invented Pandora's box?

Hesiod

The Pandora's box origin story is not absolute, but an early telling of the story can be traced back to Hesiod. Hesiod was a 7th-century farmer and writer. He mentions Pandora in two of his poems; in "Works and Days," a poem about living well, Hesiod relates the story of Pandora in lines 60-105.

What does Pandora's box mean today?

A "Pandora's box" is a metaphor in our modern languages, and the proverbial phrase refers to a source of endless complications or trouble arising from a single, simple miscalculation.

What is an example of Pandora's box?

noun. a source of extensive but unforeseen troubles or problems: The senate investigation turned out to be a Pandora's box for the administration.

What does Pandora's name mean?

All gifts

Meaning: All gifts. Pandora is a girl's name of Greek origin, meaning "all gifts." In Greek mythology, the first mortal woman, Pandora, was created beautiful and delightful by the gods, such that every man was attracted to her.

Moral of Pandora's Box Story

The story of Pandora's box warns you that curiosity is a good thing when applied to learning new things. But sometimes, leaving your curiosity unchecked, along with disobedience, can get you into dangerous situations.

How do you start building a project?

How to start a project: The 5 phases

1. Perform a feasibility study.
2. Define project goals.
3. Create a project charter.
4. Identify key stakeholders.
5. Select project management tools.

How do we start a project?

The Project Initiation Process - 7 Steps to Start Right!

1. Define Your Goals. First things first: decide what you want to achieve. ...
2. Outline The Scope. ...
3. Budgeting and Timeline. ...
4. Defining Who Is Involved And When. ...
5. Assemble the Team. ...
6. Project Kick-Off Meeting. ...
7. Tracking and Monitoring.

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How to do a small project?

Follow these steps to make mini project in college:

1. Find a simple topic that interests you.
2. Draft a clear-cut plan before you start working on your project.
3. List 3 potential topics and choose the one that fits your future career goals.
4. Always have a backup project idea & plan ready.

More items...

How small can a project be?

A small project can be defined as a short timeline – up to 6 weeks, or as having only one or two deliverables. The definition tends to be different for each organization, but the problem is the same.

How to plan small projects?

How To Write A Project Plan in 6 Steps

1. Establish Project Scope And Metrics. ...

2. Identify Key Project Stakeholders. ...
3. Outline Deliverables. ...
4. Develop Actionable Tasks. ...
5. Assign Tasks And Deadlines. ...
6. Share, Gather Feedback, And Adjust The Project Plan As Necessary.

What is an Arduino?

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Introduction

Arduino is an open-source platform used for building electronics projects. Arduino consists of both a physical programmable circuit board (often referred to as a microcontroller) and a piece of software, or IDE (Integrated Development Environment) that runs on your computer, used to write and upload computer code to the physical board.

The Arduino platform has become quite popular with people just starting out with electronics, and for good reason. Unlike most previous programmable circuit boards, the Arduino does not need a separate piece of hardware (called a programmer) in order to load new code onto the board -- you can simply use a USB cable. Additionally, the Arduino IDE uses a simplified version of C++, making it easier to learn to program. Finally, Arduino provides a standard form factor that breaks out the functions of the micro-controller into a more accessible package.

This is an Arduino Uno

The Uno is one of the more popular boards in the Arduino family and a great choice for beginners. We'll talk about what's on it and what it can do later in the tutorial.

What Does it Do?

The Arduino hardware and software was designed for artists, designers, hobbyists, hackers, newbies, and anyone interested in creating interactive objects or environments. Arduino can interact with buttons, LEDs, motors, speakers, GPS units, cameras, the internet, and even your smart-phone or your TV! This flexibility combined with the fact that the Arduino software is free, the hardware boards are pretty cheap, and both the software and hardware are easy to learn has led to a large community of users who have contributed code and released instructions for a huge variety of Arduino-based projects.

For everything from robots and a heating pad hand warming blanket to honest fortune-telling machines, and even a Dungeons and Dragons dice-throwing gauntlet, the Arduino can be used as the brains behind almost any electronics project.

_Wear your nerd cred on your sleev... err, arm. _

What's on the board?

There are many varieties of Arduino boards (explained on the next page) that can be used for different purposes. Some boards look a bit different from the one below, but most Arduinos have the majority of these components in common:

Power (USB / Barrel Jack)

Every Arduino board needs a way to be connected to a power source. The Arduino UNO can be powered from a USB cable coming from your computer or a wall power supply (like this) that is terminated in a barrel jack. In the picture above the USB connection is labeled (1) and the barrel jack is labeled (2).

The USB connection is also how you will load code onto your Arduino board. More on how to program with Arduino can be found in our [Installing and Programming Arduino](#) tutorial.

NOTE: Do NOT use a power supply greater than 20 Volts as you will overpower (and thereby destroy) your Arduino. The recommended voltage for most Arduino models is between 6 and 12 Volts.

Pins (5V, 3.3V, GND, Analog, Digital, PWM, AREF)

The pins on your Arduino are the places where you connect wires to construct a circuit (probably in conjunction with a breadboard and some wire. They usually have black plastic 'headers' that allow you to just plug a wire right into the board. The Arduino has several different kinds of pins, each of which is labeled on the board and used for different functions.

- GND (3): Short for 'Ground'. There are several GND pins on the Arduino, any of which can be used to ground your circuit.

- 5V (4) & 3.3V (5): As you might guess, the 5V pin supplies 5 volts of power, and the 3.3V pin supplies 3.3 volts of power. Most of the simple components used with the Arduino run happily off of 5 or 3.3 volts.

- Analog (6): The area of pins under the 'Analog In' label (A0 through A5 on the UNO) are Analog In pins. These pins can read the signal from an analog sensor (like a temperature sensor) and convert it into a digital value that we can read.

- Digital (7): Across from the analog pins are the digital pins (0 through 13 on the UNO). These pins can be used for both digital input (like telling if a button is pushed) and digital output (like powering an LED).

- PWM (8): You may have noticed the tilde (~) next to some of the digital pins (3, 5, 6, 9, 10, and 11 on the UNO). These pins act as normal digital pins, but can also be used for something called Pulse-Width Modulation (PWM). We have a tutorial on PWM, but for now, think of these pins as being able to simulate analog output (like fading an LED in and out).

- AREF (9): Stands for Analog Reference. Most of the time you can leave this pin alone. It is sometimes used to set an external reference voltage (between 0 and 5 Volts) as the upper limit for the analog input pins.

Reset Button

Just like the original Nintendo, the Arduino has a reset button (10). Pushing it will temporarily connect the reset pin to ground and restart any code that is loaded on the Arduino. This can be very useful if your code doesn't repeat, but you want to test it multiple times. Unlike the original Nintendo however, blowing on the Arduino doesn't usually fix any problems.

Power LED Indicator

Just beneath and to the right of the word “UNO” on your circuit board, there’s a tiny LED next to the word ‘ON’ (11). This LED should light up whenever you plug your Arduino into a power source. If this light doesn’t turn on, there’s a good chance something is wrong. Time to re-check your circuit!

TX RX LEDs

TX is short for transmit, RX is short for receive. These markings appear quite a bit in electronics to indicate the pins responsible for serial communication. In our case, there are two places on the Arduino UNO where TX and RX appear -- once by digital pins 0 and 1, and a second time next to the TX and RX indicator LEDs (12). These LEDs will give us some nice visual indications whenever our Arduino is receiving or transmitting data (like when we’re loading a new program onto the board).

Main IC

The black thing with all the metal legs is an IC, or Integrated Circuit (13). Think of it as the brains of our Arduino. The main IC on the Arduino is slightly different from board type to board type, but is usually from the ATmega line of IC’s from the ATMEL company. This can be important, as you may need to know the IC type (along with your board type) before loading up a new program from the Arduino software. This information can usually be found in writing on the top side of the IC. If you want to know more about the difference between various IC’s, reading the datasheets is often a good idea.

Voltage Regulator

The voltage regulator (14) is not actually something you can (or should) interact with on the Arduino. But it is potentially useful to know that it is there and what it’s for. The voltage regulator does exactly what it says -- it controls the amount of voltage that is let into the Arduino board. Think of it as a kind of gatekeeper; it will turn away an extra voltage that might harm the circuit. Of course, it has its limits, so don’t hook up your Arduino to anything greater than 20 volts.

This is a screenshot of the Arduino IDE.

Believe it or not, those 10 lines of code are all you need to blink the on-board LED on your Arduino. The code might not make perfect sense right now, but, after reading this tutorial and the many more Arduino tutorials waiting for you on our site, we’ll get you up to speed in no time!

The Arduino Family

Arduino makes several different boards, each with different capabilities. In addition, part of being open source hardware means that others can modify and produce derivatives of Arduino boards that provide even more form factors and functionality. If you’re not sure which one is right for your project, check this guide for some helpful hints. Here are a few options that are well-suited to someone new to the world of Arduino:

Arduino Uno (R3)

The Uno is a great choice for your first Arduino. It’s got everything you need to get started, and nothing you don’t. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a USB connection, a power jack, a reset button and more. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power

it with a AC-to-DC adapter or battery to get started.

LilyPad Arduino

This is LilyPad Arduino main board! LilyPad is a wearable e-textile technology developed by Leah Buechley and cooperatively designed by Leah and SparkFun. Each LilyPad was creatively designed with large connecting pads and a flat back to allow them to be sewn into clothing with conductive thread. The LilyPad also has its own family of input, output, power, and sensor boards that are also built specifically for e-textiles. They're even washable!

RedBoard

At SparkFun we use many Arduinos and we're always looking for the simplest, most stable one. Each board is a bit different and no one board has everything we want -- so we decided to make our own version that combines all our favorite features.

The RedBoard can be programmed over a USB Mini-B cable using the Arduino IDE. It'll work on Windows 8 without having to change your security settings (we used signed drivers, unlike the UNO). It's more stable due to the USB/FTDI chip we used, plus it's completely flat on the back, making it easier to embed in your projects. Just plug in the board, select "Arduino UNO" from the board menu and you're ready to upload code. You can power the RedBoard over USB or through the barrel jack. The on-board power regulator can handle anything from 7 to 15VDC.

Arduino Mega (R3)

The Arduino Mega is like the UNO's big brother. It has lots (54!) of digital input/output pins (14 can be used as PWM outputs), 16 analog inputs, a USB connection, a power jack, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. The large number of pins make this board very handy for projects that require a bunch of digital inputs or outputs (like lots of LEDs or buttons).

Arduino Leonardo

The Leonardo is Arduino's first development board to use one microcontroller with built-in USB. This means that it can be cheaper and simpler. Also, because the board is handling USB directly, code libraries are available which allow the board to emulate a computer keyboard, mouse, and more!

The Extended Family

While your Arduino board sure is pretty, it can't do a whole lot on its own -- you've got to hook it up to something. There are lots of tutorials here on learn as well as the links back in the 'What does it do' section, but rarely do we talk about the general kinds of things you can easily hook into. In this section we'll introduce basic sensors as well as Arduino shields, two of the most handy tools to use in bringing your projects to life.

Sensors

With some simple code, the Arduino can control and interact with a wide variety of sensors - things that can measure light, temperature, degree of flex, pressure, proximity, acceleration, carbon monoxide, radioactivity, humidity, barometric pressure, you name it, you can sense it!

Just a few of the sensors that are easily compatible with Arduino

Shields

Additionally, there are these things called shields -- basically they are pre-built circuit boards that fit on top of your Arduino and provide additional capabilities -- controlling motors, connecting to the internet, providing cellular or other wireless communication, controlling an LCD screen, and much more.

A partial selection of available shields to extend the power of your Arduino